

Trainer Debrief — Lab 3

DMAIC at Depth — Recap, $Y = f(X)$ and Baseline Sigma for Your Project

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Re-derive the DMAIC roadmap and express your capstone as $Y = f(X)$ with a baseline sigma level (K1, A4).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-03-baseline-sigma-calculator.xlsx

Full-year totals: 49,565 units, 1,742 defects, 1,442 defective units. $DPU = 1742/49565 = 0.0351$. $DPO = 0.0351/4 = 0.00878$. $DPMO = 8,784$. Yield (units with no defect) = 97.1%, so the defect rate is 2.9% — the baseline every later lab quotes. From the DPMO table that is about 3.9 sigma at the OPPORTUNITY level, but only ~2.9 sigma at the UNIT level; always state which one you mean. $RTY = 0.980 \times 0.971 \times 0.991 \times 0.996 \times 0.971 \times 0.997 = 0.909$ — a 9.1% hidden factory that the final-yield number completely conceals. The seal weld and burst test steps are the two worst, which is exactly where the project is aimed.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

You can explain all five DMAIC tollgates without notes, your Y is a measurable customer-facing output with at least twelve classified Xs, and your baseline DPMO and sigma level are calculated from real counts.

Where this sits in the capstone

A thorough recap of the DMAIC roadmap — not a skim, because you must be able to mentor Green Belts through it. Then apply it: express your capstone problem as $Y = f(X)$, and convert your baseline defect data into DPMO and a sigma level so improvement can be proven numerically at the end. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27

Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30
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If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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